4.6 Extraction
4.6.1 If the sample was not acidified at time of collection, add 5 ml hydrochloric acid (4.5.1) to the sample bottle. After mixing the sample, check the pH by touching pH-sensitive ,paper to the cap to ensure that the pH is 2, or lower. Add more acid if necessax-y.
4.6.2 Pour the sample into a separatory funnel and add 5 g of sodium chloride.
4.6.3 Add 50 ml carbon tetrachloride to the sample . bottle. Cap tightly and thoroughly shake the bottle to rinse the inside and cap. rransfer the solvent into the separatory funnel and extract by shaking vigorously for two minutes. Allow the layers to separate.
4.6.4 Drain the solvent layer through a funnel containing solvent-moistened filter paper into a 100 ml volumetric flask.
4.6.5 Repeat steps 4.6.3 and 4.6.4 with an additional 50 ml portion of fresh solvent; combine all solvent in the volumetric flask.
4.6.6 Rinse the
small portions of
volumetric flask.
flask. Mix well.
top of the separatorY ,funnel, filter paper and funnel with

---CHUNK---

SOURCE: MEPC.60(33).pdf
solvent; combine all solvent in the volumetric flask.
4.6.6 Rinse the
small portions of
volumetric flask.
flask. Mix well.
top of the separatorY ,funnel, filter paper and funnel with
carbon tet,rachloride . and collect the rinsings in the
Adjust the e~, tract volume up to 100 ml and stopper the
4.6.1 Drain the water layer into a 1,000 mI. gradua,~ed cylinder and estimate the sample volume to the nearest 5 mI.
4.1 Infrared Spectroscopy
4.1.1 Prepare the infrared spectrophotometer according to manufacturer's instructions.
"
4.1.2 Rinse a cell with two volumes of solution to be measured, then completely fill the cell with solution. l'lace a mat;.ched cell containing carbon tetrachloride in the reference beam.
4.1.3 Scan samples and standards from 3,000 cm- l to 2,100 cm- 1 .
NOTE 1: Single beam and non.,..scaning spectrophotometers can be used for this test. Follow the manufacturer's instructions and measure the absorbance directly at O~ near 2,930 em-I.
W/9146D/EWP
)
Resolution MEPC.60(33)
adopted on 30 October 1992
GUIDELINES AND SPECIFICATIONS FOR POLLUTION PREVENTION EQUIPMENT FOR MACHINERY SPACE BILGES OF SHIPS)
W/9146D/EWP
MEPC 33/20/Add.1
ANNEX 10

---CHUNK---

SOURCE: MEPC.60(33).pdf
directly at O~ near 2,930 em-I.
W/9146D/EWP
)
Resolution MEPC.60(33)
adopted on 30 October 1992
GUIDELINES AND SPECIFICATIONS FOR POLLUTION PREVENTION
EQUIPMENT FOR MACHINERY SPACE BILGES OF SHIPS)
W/9146D/EWP
MEPC 33/20/Add.1
ANNEX 10
Page 23
Figure 4 - Speotrum illustrating baseline construction
o~---~~---r----~----~----~----r-----~----'
0.1
0.2
0.3
0.4
0.7
0.8
0.9
1.0 L-....---.J __ .J... __ .L.... __ .L...._.....I"---~-:----'------'
. 3000
WAVENUMBER leM" 'I
3200
Resolution MEPC.60(33)
adopted on 30 October 1992
GUIDELINES AND SPECIFICATIONS FOR POLLUTION PREVENTION
EQUIPMENT FOR MACHINERY SPACE BILGES OF SHIPSMEPC 33/20/Add.1
ANNEX 10
Page 24
4.7.4 Construct a straight ba,seline un,der the hydro .carhon , band as illustrated in figure 4. If the scan is recorded on abosorbance paper, read the absorbance of the peak maximum at 2,930 cm- 1 and subtract the absorbance of the baseline at that point. If the scan is recorded on transmittance paper, the net absorbance is:
\,T (base line)
%T (peak maximum) ·
4.7.5 Prepare a calibration plot of net absorbance vs mg/lOO ml oil using the response of the standards.

---CHUNK---

SOURCE: MEPC.60(33).pdf
the baseline at that point. If the scan is recorded on transmittance paper, the net absorbance is:
\,T (base line)
%T (peak maximum) ·
4.7.5 Prepare a calibration plot of net absorbance vs mg/lOO ml oil using the response of the standards.
NOTE 2: The oil concentration may b~ plotted as per cent of stock standard. When this procedure is used, the concentration of the stock standard must be

used in the calculations (4.8.2).
4.7.6 If the net absorbance of a sample exceeds 0.8 or the linear range of the instrument as determined by the calibration plot, prepare a dilution of the sample by pipetting an appropriate volume of the extract into a volumetric flask and diluting to volume.' If the absorbance is less than 0.1, more accurate results can be obtained by using a longer path-length cell.
4.8 Calculations
4.8.1 Use th~ calibration plot to calculate the mg of oil in each 100 ml of sample extract or dilution.
4.8.2 Calculate the oil content ln the sample using the formula:
where:
mg/l oil = E x D x 1000
V
R = mg of oil in 100 ml solution (determined from calibration plot)
D = extract dilution factor, if used (4.7.6)
V = volume of sample, in millilitres (4.6.7)

---CHUNK---

SOURCE: MEPC.60(33).pdf
where:
mg/l oil = E x D x 1000
V
R = mg of oil in 100 ml solution (determined from calibration plot)
D = extract dilution factor, if used (4.7.6)
V = volume of sample, in millilitres (4.6.7)
4.8.3 Report results to two significant figures for levels below 100 mg/l.
NOTE 3: For quality control, a reagent blank should be carried through each step of the procedure.
4.8.4 For purposes of comparison to meter records, the results should also be presented in parts per million (volume/volume) with due allowance for the relative density of the o.i .l. , J
PART 5 - DOCUMENTATION OF APPROVAL
5.1 Certificate of Type Approval for pollution prevention equipment
W/9146D/EWP
)
Resolution MEPC.60(33)
adopted on 30 October 1992
GUIDELINES AND SPECIFICATIONS FOR POLLUTION PREVENTION EQUIPMENT FOR MACHINERY SPACE BILGES OF SHIPS)
MEPC 33/20/Add.l
ANNEX 10
Page 25
5.1.1 Satisfactory compliance with all the test requirements enumerated in parts 1 and 2 of this annex should be shown in the Certificate of Type Approval issued by the Administration in the format specified in paragraph 5.1.2 below. An Administration may issue a Certificate of Type

---CHUNK---

SOURCE: MEPC.60(33).pdf
in parts 1 and 2 of this annex should be shown in the Certificate of Type Approval issued by the Administration in the format specified in paragraph 5.1.2 below. An Administration may issue a Certificate of Type

Approval based on separate testing or on testing already carried out under supervision by another ,Administration.
5.1.2 A Certificate of Type Approval should be in the format shown in appendix 1 or 2 to this annex. The Certificate should identify the type and model of the pollution p'revention equipment to which it applies and identify equipment assembly drawings, duly dated. Each drawing should bear the model specification numbers Or equivalent identification details. The Certificate should include the full performance test protocol on ' which it is based. If a Certificate of Type Approval is issued by an Administration based on a Certificate previously issued by another Administration, the Certificate should identify the Administration which conducted the tests on the pollution prevention equipment and a copy of the original test results should be attached to it.
5.2 Format of environmental test protocol

---CHUNK---

SOURCE: MEPC.60(33).pdf
should identify the Administration which conducted the tests on the pollution prevention equipment and a copy of the original test results should be attached to it.
5.2 Format of environmental test protocol
5.2.1 Satisfactory compliance with the environmental tests laid down in